

Step 1 — Install spaCy and Download Model

```
# Install spaCy
!pip install spacy

# Download English language model
!python -m spacy download en_core_web_sm
```

```
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.11)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.11)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.0)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.10)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.3)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.2)
Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (8.3.4)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.10.1)
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.4.3)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.6)
Requirement already satisfied: weasel<0.5.0,>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.4.2)
Requirement already satisfied: typer-slim<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.3.0)
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (4.67.1)
Requirement already satisfied: numpy>=1.19.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.26.4)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.32.3)
Requirement already satisfied: pydantic!=1.8,!<1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.10.6)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.1.4)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from spacy) (75.8.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (24.2)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!<1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy)) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!<1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy)) (2.41.4)
Requirement already satisfied: typing-extensions>=4.14.1 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!<1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy)) (4.14.1)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!<1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy)) (0.4.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy)) (3.4.1)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy)) (3.10.1)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy)) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy)) (2025.11.12)
Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy)) (1.3.0)
Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy)) (0.0.1)
Requirement already satisfied: typer>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy)) (0.24.0)
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy)) (0.7.0)
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy)) (5.2.1)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2 in /usr/local/lib/python3.12/dist-packages (from spacy)) (3.0.2)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart-open<8.0.0,>=5.2.1 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy)) (1.17.0)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy)) (8.2.1)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy)) (1.3.0)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy)) (12.3.0)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy)) (0.0.2)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy)) (2.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy)) (2.13.0)
Requirement already satisfied: mdurl~>0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy)) (0.1.2)
Collecting en-core-web-sm==3.8.0
```

Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0.tar.gz 12.8/12.8 MB 92.8 MB/s eta 0:00:00

✓ Download and installation successful

You can now load the package via `spacy.load('en_core_web_sm')`

⚠ Restart to reload dependencies

If you are in a Jupyter or Colab notebook, you may need to restart Python in order to load all the package's dependencies. You can do this by selecting the 'Restart kernel' or 'Restart runtime' option.

Step 2 — Import Libraries and Load Model

```
import spacy
import pandas as pd
from spacy import displacy

# Load spaCy English model
```

```
nlp = spacy.load("en_core_web_sm")

print("spaCy model loaded successfully")

spaCy model loaded successfully
```

Step 3 — Prepare Dataset (15 Sentences)

```
sentences = [
    "The cat chased the mouse.",
    "She reads a book every day.",
    "The teacher explained the lesson clearly.",
    "The boy kicked the ball and ran away.",
    "I went to the store because I needed milk.",
    "The scientist who discovered the vaccine won an award.",
    "Did you complete the assignment?",
    "Why are students studying late tonight?",
    "The project was completed by the team.",
    "The cake was baked by my mother.",
    "Technology improves our daily lives.",
    "Social media connects people around the world.",
    "The programmer fixed the bug quickly.",
    "Artificial intelligence is transforming industries.",
    "The laptop on the table belongs to John."
]
```

Step 4 — Process Sentences Using spaCy

This step performs tokenization, POS tagging, and dependency parsing.

```
docs = []

for sentence in sentences:
    doc = nlp(sentence)
    docs.append(doc)

print("\nSentence:", sentence)
for token in doc:
    print(token.text, token.pos_, token.dep_)
```

```
Sentence: The cat chased the mouse.
The DET det
cat NOUN nsubj
chased VERB ROOT
the DET det
mouse NOUN dobj
. PUNCT punct
```

```
Sentence: She reads a book every day.
She PRON nsubj
reads VERB ROOT
a DET det
book NOUN dobj
every DET det
day NOUN npadvmod
. PUNCT punct
```

```
Sentence: The teacher explained the lesson clearly.
The DET det
teacher NOUN nsubj
explained VERB ROOT
the DET det
lesson NOUN dobj
clearly ADV advmod
. PUNCT punct
```

Sentence: The boy kicked the ball and ran away.

The DET det
 boy NOUN nsubj
 kicked VERB ROOT
 the DET det
 ball NOUN dobj
 and CCONJ cc
 ran VERB conj
 away ADV advmod
 . PUNCT punct

Sentence: I went to the store because I needed milk.

I PRON nsubj
 went VERB ROOT
 to ADP prep
 the DET det
 store NOUN pobj
 because SCONJ mark
 I PRON nsubj
 needed VERB advcl
 milk NOUN dobj
 . PUNCT punct

Sentence: The scientist who discovered the vaccine won an award.

The DET det
 scientist NOUN nsubj
 who PRON nsubj
 discovered VERB relcl
 the DET det
 vaccine NOUN dobj
 won VERB ROOT

Step 5 — Extract Dependency Relations

This will create a table showing token, lemma, POS tag, dependency, and head word.

```
data = []

for doc in docs:
    for token in doc:
        data.append([
            doc.text,
            token.text,
            token.lemma_,
            token.pos_,
            token.dep_,
            token.head.text
        ])

df = pd.DataFrame(data, columns=[
    "Sentence",
    "Token",
    "Lemma",
    "POS",
    "Dependency",
    "Head Word"
])

df
```

	Sentence	Token	Lemma	POS	Dependency	Head Word	
0	The cat chased the mouse.	The	the	DET	det	cat	
1	The cat chased the mouse.	cat	cat	NOUN	nsubj	chased	
2	The cat chased the mouse.	chased	chase	VERB	ROOT	chased	
3	The cat chased the mouse.	the	the	DET	det	mouse	
4	The cat chased the mouse.	mouse	mouse	NOUN	dobj	chased	
...	...	...	...	...	...	...	
109	The laptop on the table belongs to John.	table	table	NOUN	pobj	on	
110	The laptop on the table belongs to John.	belongs	belong	VERB	ROOT	belongs	
111	The laptop on the table belongs to John.	to	to	ADP	prep	belongs	
112	The laptop on the table belongs to John.	John	John	PROPN	pobj	to	
113	The laptop on the table belongs to John.	.	.	PUNCT	punct	belongs	

114 rows × 6 columns

Next steps:

[Generate code with df](#)[New interactive sheet](#)

Step 6 — Identify Sentence Structure (Subject, Verb, Object)

```

for doc in docs:
    print("\nSentence:", doc.text)

    subject = ""
    verb = ""
    obj = ""

    for token in doc:
        if token.dep_ == "nsubj":
            subject = token.text
        if token.dep_ == "ROOT":
            verb = token.text
        if token.dep_ == "dobj":
            obj = token.text

    print("Subject:", subject)
    print("Main Verb:", verb)
    print("Object:", obj)

```

Sentence: The cat chased the mouse.
 Subject: cat
 Main Verb: chased
 Object: mouse

Sentence: She reads a book every day.
 Subject: She
 Main Verb: reads
 Object: book

Sentence: The teacher explained the lesson clearly.
 Subject: teacher
 Main Verb: explained
 Object: lesson

Sentence: The boy kicked the ball and ran away.
 Subject: boy
 Main Verb: kicked
 Object: ball

Sentence: I went to the store because I needed milk.

Subject: I

Main Verb: went

Object: milk

Sentence: The scientist who discovered the vaccine won an award.

Subject: who

Main Verb: won

Object: award

Sentence: Did you complete the assignment?

Subject: you

Main Verb: complete

Object: assignment

Sentence: Why are students studying late tonight?

Subject: students

Main Verb: are

Object:

Sentence: The project was completed by the team.

Subject:

Main Verb: completed

Object:

Sentence: The cake was baked by my mother.

Subject:

Main Verb: baked

Object:

Sentence: Technology improves our daily lives.

Subject: Technology

Main Verb: improves

Object: lives

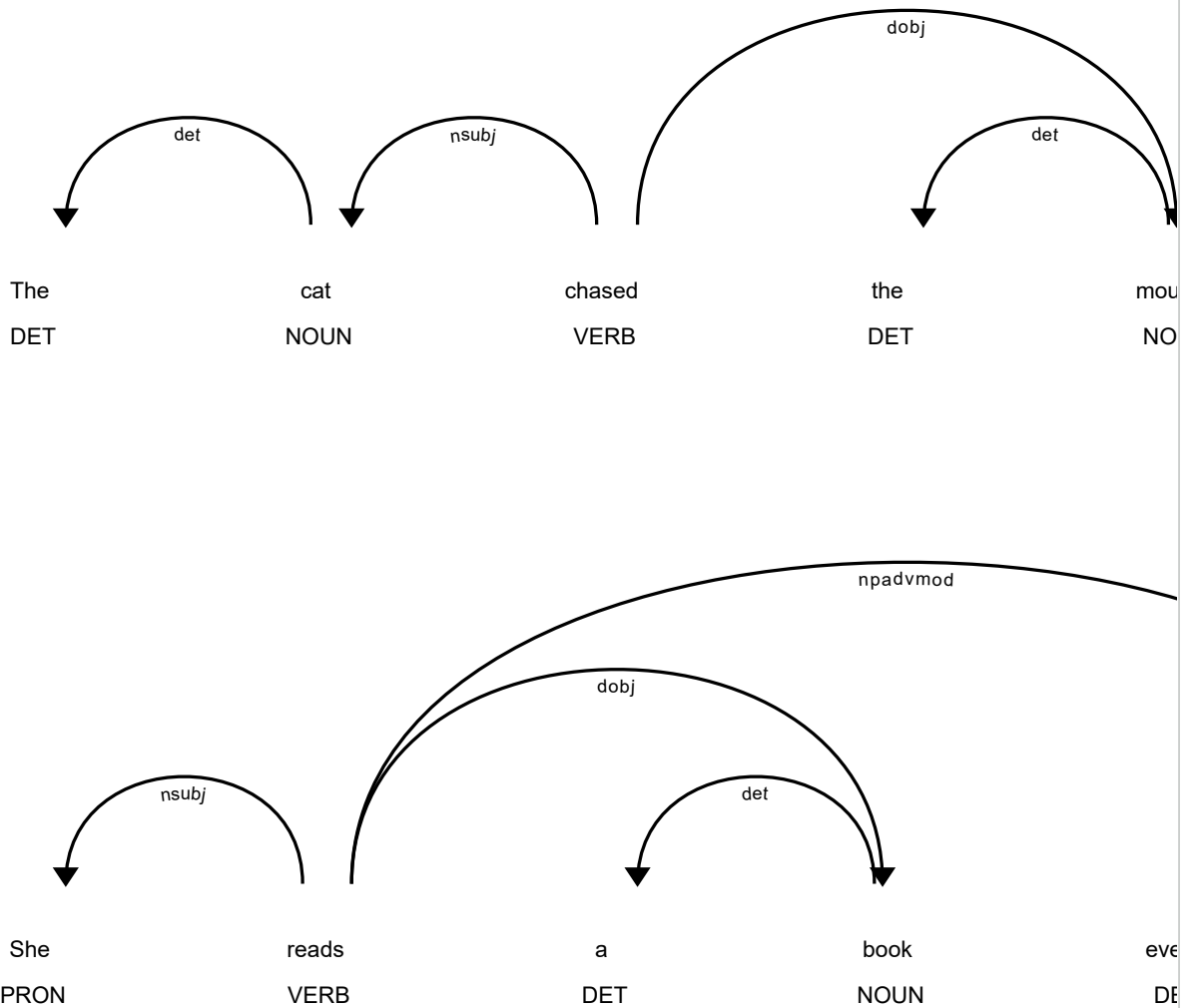
Sentence: Social media connects people around the world.

Subject: media

Step 7 — Dependency Tree Visualization

This will display dependency trees for 5 sentences.

```
for i in range(5):  
    doc = nlp(sentences[i])  
    displacy.render(doc, style="dep", jupyter=True)
```

Step 8 — Show Named Dependency Relations (Optional Check)

```
for doc in docs:
    print("\nSentence:", doc.text)

    for token in doc:
        print(f"{token.text} --> {token.dep_} --> {token.head.text}")
```

```
Sentence: The cat chased the mouse.
The --> det --> cat
cat --> nsubj --> chased
chased --> ROOT --> chased
the --> det --> mouse
mouse --> doobj --> chased
. --> punct --> chased
```

```
Sentence: She reads a book every day.
She --> nsubj --> reads
reads --> ROOT --> reads
a --> det --> book
book --> dobj --> reads
every --> det --> day
day --> npadvmod --> reads
. --> punct --> reads
```

Sentence: The teacher explained the lesson clearly.

The --> det --> teacher

teacher --> nsubj --> explained

explained --> ROOT --> explained

the --> det --> lesson

lesson --> dobj --> explained

clearly --> advmod --> explained

. --> punct --> explained

det

Sentence: The boy kicked the ball and ran away.

The --> det --> boy

boy --> nsubj --> kicked

kicked --> ROOT --> kicked

the --> det --> ball

ball --> dobj --> kicked

and --> cc --> kicked

ran --> conj --> kicked

away --> advmod --> ran

. --> punct --> kicked

Sentence: I went to the store because I needed milk.

I --> nsubj --> went

went --> ROOT --> went

to --> prep --> went

the --> det --> store

store --> pobj --> to

because --> mark --> needed

I --> nsubj --> needed

.